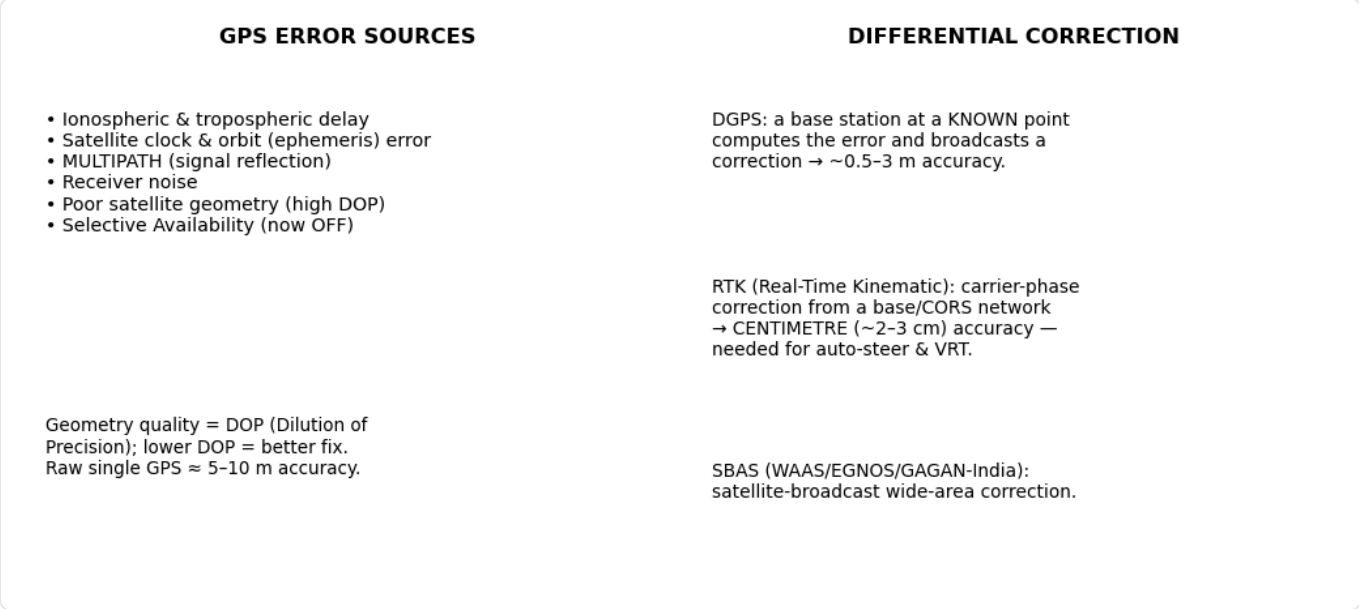
UNIT 10 · TOPIC 3(B)

GPS Errors, Correction Methods and GNSS Constellations

Syllabus topic: GPS basics: space, receiver and control segments; error and correction; function and usage

Error sources



GPS error sources and differential correction

- IONOSPHERIC and TROPOSPHERIC signal delays.
- SATELLITE CLOCK and ORBIT (ephemeris) errors.
- MULTIPATH — signals reflecting off objects or the ground.
- Receiver noise.
- Poor satellite GEOMETRY, measured by **DOP** (Dilution of Precision — lower is better).
- Selective Availability (deliberate degradation) was switched off in **2000**.

Raw single-receiver (autonomous) GPS gives about **5–10 m** — too coarse for row-crop operations.

Correction methods

Method	How	Accuracy
DGPS (Differential GPS)	A base receiver at a precisely KNOWN point computes the current error and broadcasts a correction to nearby rovers	Sub-metre to ~3 m
RTK (Real-Time Kinematic)	Uses the satellite CARRIER PHASE with corrections from a local base station or a CORS network	CENTIMETRE (~2–3 cm) — needed for auto-steer and precise VRT
SBAS (Satellite-Based Augmentation)	Wide-area corrections broadcast from geostationary satellites — WAAS (USA), EGNOS (Europe), GAGAN (India)	Wide-area improvement

GNSS constellations

System	Operator	Type
GPS	USA	Global (~31 satellites)
GLONASS	Russia	Global
GALILEO	European Union	Global
BeiDou (BDS)	China	Global
NavIC / IRNSS	India (ISRO)	Regional (7 satellites)
QZSS	Japan	Regional

NavIC (Navigation with Indian Constellation)

- **7 satellites:** 3 geostationary (GSO) + 4 inclined geosynchronous, plus a ground control/timing/ranging network.
- Covers India and up to **1,500 km** beyond its borders.
- Positioning accuracy better than **10 m** over India (~**20 m** in the surrounding region).
- Two services: **SPS** (Standard, open) and **RS** (Restricted, encrypted, authorised users).
- NavIC signals interoperate with GPS/GLONASS/Galileo/BeiDou. India's SBAS is **GAGAN** (GPS-Aided GEO Augmented Navigation).

Status update (October 2026): 7 satellites is the DESIGN constellation and the exam answer. In practice atomic-clock failures (most recently IRNSS-1F, March 2026) and the stranding of NVS-02 in transfer orbit (2025) left only 3 satellites providing navigation in 2026 — below the 4 needed for standalone positioning (timing still works). NVS-03, carrying an indigenous rubidium clock, was being readied for launch around mid-October 2026 to restore the minimum. Check ISRO for the latest.

> Exam tip: NavIC = 7 satellites, < 10 m, India + 1500 km, SPS/RS; RTK ≈ 2–3 cm; ≥ 4 satellites for a fix.

Practice MCQs

Q21. Which is a GPS error source?

(a) Ionospheric delay
(b) Multipath
(c) Clock/orbit error
(d) All of these

Q22. Multipath error is caused by:

(a) Signal reflections off objects/ground
(b) Battery drain
(c) Wind
(d) Rain only

Q23. Autonomous (single-receiver) GPS accuracy is about:

(a) 1 mm
(b) 5–10 m
(c) 100 m
(d) 1 km

Q24. Differential GPS (DGPS) uses a base station at a ___ point to broadcast corrections:

(a) Random
(b) Known (surveyed)
(c) Moving
(d) Underground

Q25. DGPS accuracy is roughly:

(a) 50 m
(b) Sub-metre to ~3 m
(c) 500 m
(d) 1 mm

Q26. Centimetre-level accuracy for auto-steer needs:

(a) Autonomous GPS (b) RTK (carrier-phase)
(c) A compass (d) DOP=10

Q27. RTK accuracy is approximately:

(a) 2–3 cm (b) 2–3 m
(c) 20 m (d) 200 m

Q28. India's satellite-based augmentation system (SBAS) is:

(a) WAAS (b) EGNOS
(c) GAGAN (d) MSAS

Q29. Selective Availability (deliberate GPS degradation) was:

(a) Still on (b) Turned off in 2000
(c) Never used (d) Indian only

Q30. India's regional navigation system is:

(a) GLONASS (b) NavIC / IRNSS
(c) Galileo (d) BeiDou

Q31. NavIC consists of how many satellites?

(a) 3 (b) 7
(c) 24 (d) 31

Q32. NavIC positioning accuracy over India is better than:

(a) 1 mm (b) 10 m
(c) 100 m (d) 1 km

Q33. Which is a GLOBAL GNSS operated by the European Union?

(a) BeiDou (b) Galileo
(c) GLONASS (d) QZSS

Q34. China's global navigation system is:

(a) BeiDou (b) NavIC
(c) GAGAN (d) QZSS

Answers

Q21 — (d) All of these
Q22 — (a) Signal reflections off objects/ground
Q23 — (b) 5–10 m — *Autonomous ≈ 5–10 m.*
Q24 — (b) Known (surveyed)
Q25 — (b) Sub-metre to ~3 m — *DGPS sub-metre to 3 m.*
Q26 — (b) RTK (carrier-phase) — *RTK carrier phase ≈ 2–3 cm (auto-steer).*
Q27 — (a) 2–3 cm — *RTK ≈ 2–3 cm.*
Q28 — (c) GAGAN — *SBAS = GAGAN in India.*
Q29 — (b) Turned off in 2000
Q30 — (b) NavIC / IRNSS — *NavIC/IRNSS = India.*
Q31 — (b) 7 — *7 satellites in the design constellation (see the 2026 status note in Topic 3b).*
Q32 — (b) 10 m — *< 10 m over India.*
Q33 — (b) Galileo — *Galileo — European Union.*
Q34 — (a) BeiDou — *BeiDou — China.*

Sources: Srinivasan — Handbook of Precision Agriculture; NIIT/ICAR precision-agriculture modules; NovAtel — Introduction to GNSS; ISRO/PIB NavIC data (as given in the source notes, July 2026; operational status updated October 2026); NPTEL

precision-agriculture lectures.
